AudioReach LA Customizations

Qualcomm Technologies, Inc.

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80-VN500-11 Rev. B

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Revision History

Revision	Date	Description
A	November 2019	Initial release
B	September 2020	Terminology updates for AudioReach

Note: There is no Rev. I, O, Q, S, X, or Z per Mil. standards.

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LA Platform

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TinyALSA – Card-defs.xml

- Sound card ID and Name

```
<card>
   <id>100</id>
   <name>sm8150tavilsndc</name>
....
....
</card>
```

  - ID – Sound card number that the Qualcomm audio library (QAL) uses to open the PCM/compress sessions
  - Name – Should match the name listed in the machine driver. This is required to enumerate virtual/hardware mixer controls against the primary sound card.

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TinyALSA – Card-defs.xml (cont.)

- Add new PCM/compress devices by adding new entries for enabling concurrent PCM or compress sessions
  - ID – New ID for the PCM/compress device
  - Name – New name for the PCM /compress device
  - Properties
    - Playback – Enables playback capability
    - Capture – Enables capture capability
    - Hostless – Enables hostless capability

```
<card>
  <pcm-device>
    <id>123</id>
    <name>PCM123</name>
    <pcm_plugin>
      <so-name>libagm_pcm_plugin.so</so-name>
    </pcm_plugin>
    <props>
      <playback>1</playback>
      <capture>0</capture>
      <hostless>1</hostless>
    </props>
  </pcm-device>
....
....
  <compress-device>
    <id>105</id>
    <name>COMPRESS105</name>
    <compress_plugin>
      <so-name>libagm_compress_plugin.so</so-name>
    </compress_plugin>
    <props>
      <playback>1</playback>
      <capture>1</capture>
      <hostless>0</hostless>
    </props>
  </compress-device>
....
....
</card>
```

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QAL – Resource_manager.xml

- Resource_manager.xml contains:
  - Device names and capabilities (sample rates, channels, bit width, etc.)
  - Device (Rx/Tx) to back-end mapping
  - Sound device names

```xml
<resource_manager_info>
        <device name="QAL_DEVICE_OUT_SPEAKER" hw_intf="slim-0-rx" snd_device_name="speaker" back_end_name="SLIM-DEV1-RX-0"/>
        <device name="QAL_DEVICE_OUT_WIRED_HEADPHONE" hw_intf="slim-0-rx" snd_device_name="headphones" back_end_name="SLIM-DEV1-RX-6"/>
        <device name="QAL_DEVICE_IN_HANDSET_MIC" hw_intf="slim-0-tx" snd_device_name="handset-mic" back_end_name="SLIM-DEV1-TX-0"/>
        <device name="QAL_DEVICE_IN_SPEAKER_MIC" hw_intf="slim-0-tx" snd_device_name="handset-mic" back_end_name="SLIM-DEV1-TX-0"/>
        <device name="QAL_DEVICE_IN_TRI_MIC" hw_intf="slim-0-tx" snd_device_name="speaker-tmic" back_end_name="SLIM-DEV1-TX-0"/>
</resource_manager_info>
```

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QAL – Usecase_info.xml

- Contains use case mapping to graph KeyVectors and cal keys/tags
- QAL retrieves GKV/CKV based on the use case name and stream/device attributes
- Enables addition of new use cases by updating this xml file
- All changes are constrained to cata (config files and QACT™)

```xml
<usecase>
 <name>compress-offload</name>
  <stream-attributes>
   <pp>sa+</pp>
 </stream-attributes>
 <device-attributes>
   <bitwidth>16</bitwidth>
 </device-attributes>
 <graph_key_vector>
  <stream>
   <param key="STREAMRX" value="COMPRESS_OFFLOAD_PLAYBACK"/>
   <param key="INSTANCE" value="INSTANCE_1"/>
  </stream>
  <devicepp>
   <param key="DEVICEPP_RX" value="MBDRC" />
  </devicepp>
  <device>
   <param key="DEVICERX" value="SPEAKER" />
  </device>
 </graph_key_vector>
 <cal_key_vector>
  <key>SAMPLINGRATE</key>
  <key>CHANNELS</key>
  <key>BITWIDTH</key>
 </cal_key_vector>
 <tags>DEVICE_MFC, STREAM_MFC, MUTE,VOLUME</tags>
</usecase>
```

```xml
<usecase>
 <name>compress-offload</name>
  <stream-attributes>
  </stream-attributes>
 <device-attributes>
   <bitwidth>32</bitwidth>
 </device-attributes>
 <graph_key_vector>
  <stream>
   <param key="STREAMRX" value="COMPRESS_OFFLOAD_PLAYBACK_32"/>
   <param key="INSTANCE" value="INSTANCE_1"/>
  </stream>
  <devicepp>
  </devicepp>
  <device>
   <param key="DEVICERX" value="SPEAKER" />
  </device>
 </graph_key_vector>
 <cal_key_vector>
  <key>SAMPLINGRATE</key>
  <key>CHANNELS</key>
  <key>BITWIDTH</key>
 </cal_key_vector>
 <tags>DEVICE_MFC, STREAM_MFC, MUTE,VOLUME</tags>
</usecase>
```


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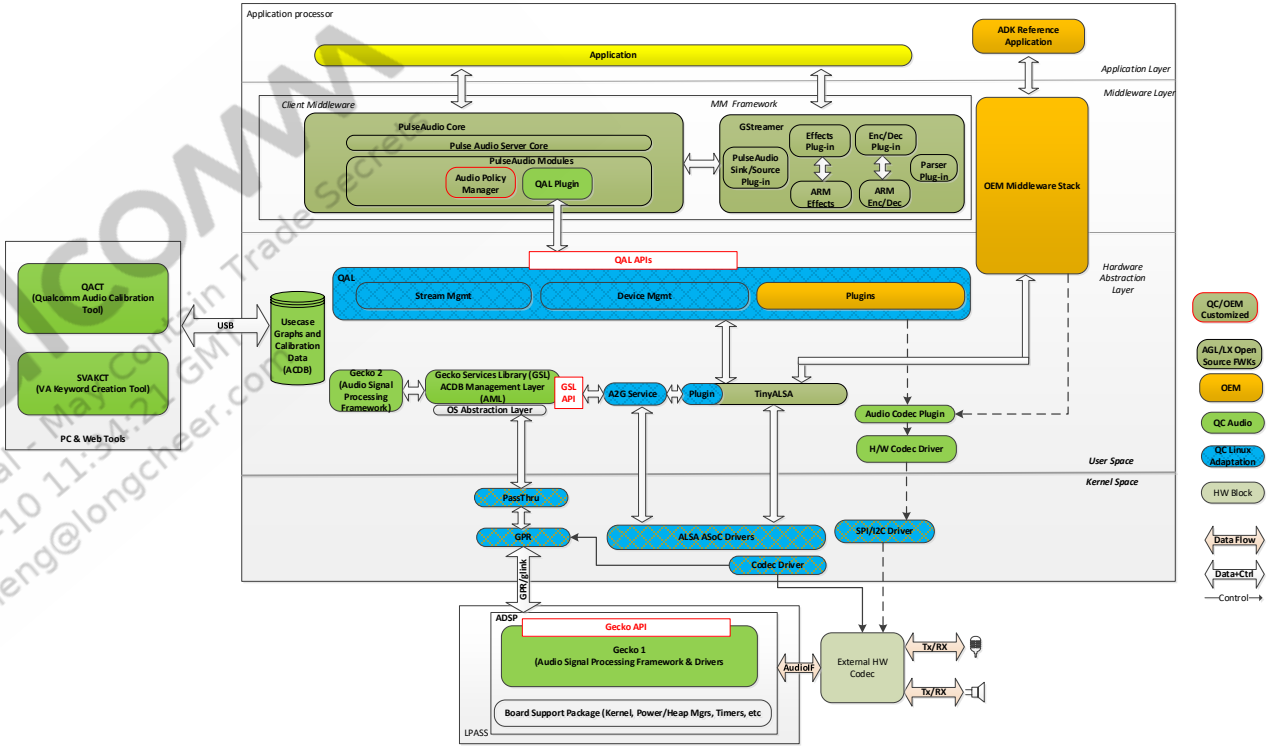
QAL – Adding Third Party (Offload) Encoders/Decoders to the Graph

- Implement the plugin interface to QAL for a new decoder or encoder (for example, FLAC)
- QAL loads plugins
- No changes are required to QAL and no need to recompile Qualcomm-delivered code
- Ability to verify independently (using CATF) without going to the target and validating



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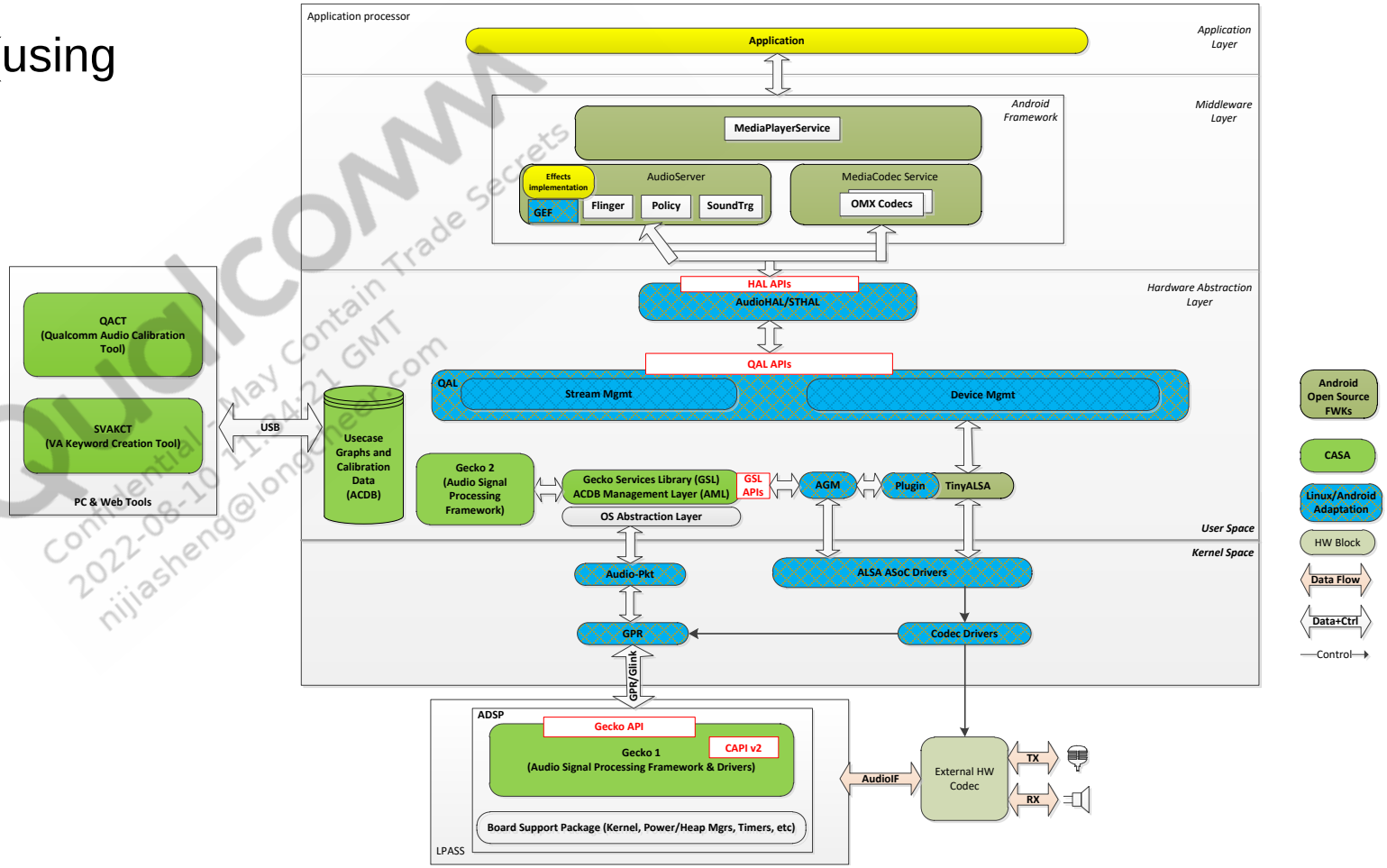
QAL – Adding Third Party PP Modules

- PP module/effects vendor writes enablement code above QAL APIs (using GEF)
- No code changes in QAL
- Use case customizations in QACT



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Questions?

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